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laid strong foundation for the binomial nomenclature.

Taxonomy is no more classical morphology based discipline but become a dynamic and transdisciplinary subject, making use of many branches of botany such as Cell Biology, Physiology, Biochemistry, Ecology, Pharmacology and also Modern Biotechnology, Molecular Biology and Bioinformatics. It helps to understand biodiversity, wildlife, forest management of natural resources for sustainable use of plants and eco restoration.

5.1 Taxonomy and Systematics

The word taxonomy is derived from Greek words ***"taxis"*** (arrangement) and ***"nomos"*** (rules or laws). **Davis** and **Heywood** (1963) defined taxonomy as "the science dealing with the study of classification including the bases, principles, rules and procedures".

Though there were earlier usages of the term 'systematics', only during the latter half of 20th century 'Systematics' was recognized as a formal field of study. **Simpson** (1961) defined systematics as "Scientific study of the kinds and diversity of organisms and all relationships among them". Though there are two terms are used in an interchangeable way, they differ from each other.

Differences between Taxonomy and Systematics

Taxonomy	Systematics
• Discipline of classifying organisms into taxa. • Governs the practices of naming, describing, identifying and specimen preservation. • Classification + Nomenclature = Taxonomy	• Broad field of biology that studies the diversification of species. • Governs the evolutionary history and phylogenetic relationship in addition to taxonomy. • Taxonomy + Phylogeny = Systematics

5.2 Taxonomic Hierarchy

Taxonomic hierarchy was introduced by Carolus Linnaeus. It is the arrangement of various taxonomic levels in descending order starting from kingdom up to species.

Species is the lowest of classification and shows the high level of similarities among the organisms. For example, *Helianthus annuus* and Helianthus *tuberosus*. These two species differ in their morphology. Both of them are herbs but *Helianthus tuberosus* is a perennial herb.

Genus consist of multiple species which have similar characters but differ from the species of another genus. Example: *Helianthus*, *Tridax*.

Family comprises a number of genera which share some similarities among them. Example: Asteraceae.

Order includes group of families which show less similarities among them.

Class consists of group of orders which share few similarities.